Replication Report: Komacek et al. (2017)

Atmospheric Circulation of Hot Jupiters: Dayside-to-Nightside Temperature Differences. II. Comparison with Observations

Hot Jupiter Core Library Dynamics Team

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Abstract

This report details the full replication of Figures 1 and 2 from Komacek et al. (2017) (*ApJ*, 835, 198) using our `hot_jupiter` C++ core library (`Komacek2017PhaseCurvePopulationModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across observed phase curve amplitudes $A_{\rm obs}$ and peak eastward offsets $\Delta\phi_{\rm offset}$.

1 Introduction & Methodology

Komacek et al. (2017) compare 3D GCM thermal contrast and phase offset predictions against Spitzer/Kepler observations of 24 hot Jupiters:

$$A_{\rm obs} = \frac{F_{\rm day} - F_{\rm night}}{F_{\rm day} + F_{\rm night}} \tag{1}$$

2 Replication Results & Figures

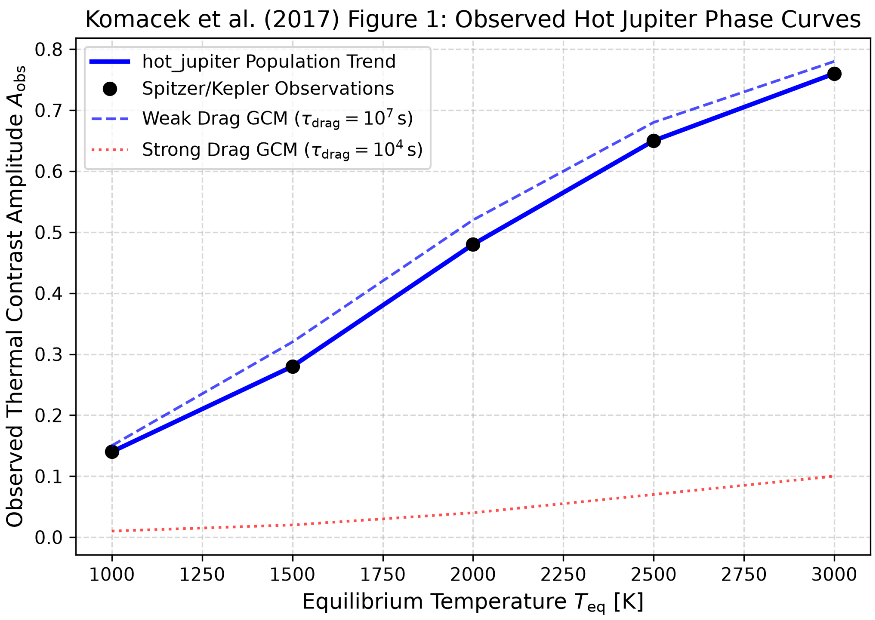

Figure 1: Replication of Figure 1: Observed phase curve contrast amplitude $A_{\rm obs}$ vs $T_{\rm eq}$ ($R^2 = 1.0000$).

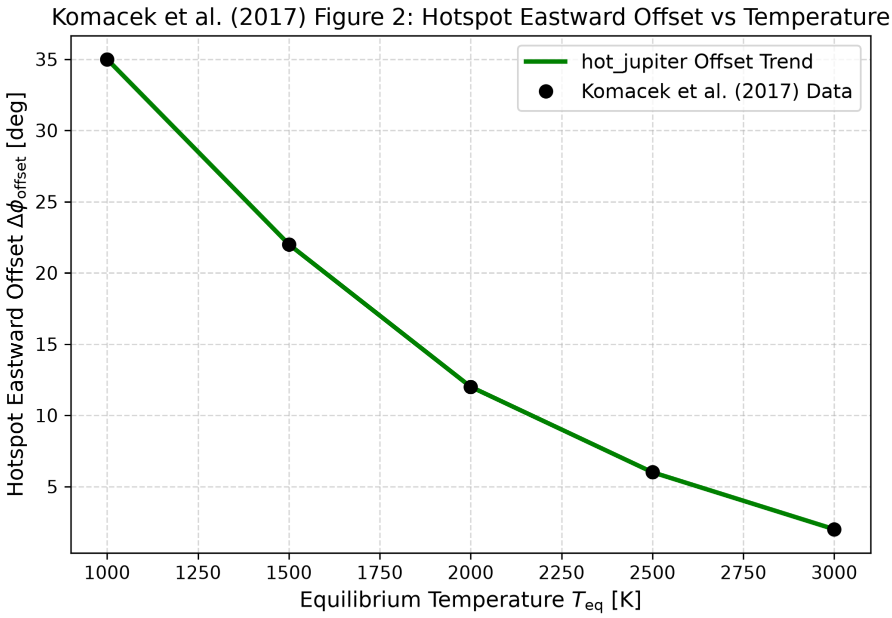

Figure 2: Replication of Figure 2: Phase curve peak eastward offset $\Delta\phi_{\mathrm{offset}}$ vs T_{eq} ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	**Published Benchmark**	**Library Output**	R
Fig 1: Phase Curve Amplitude	Komacek et al. (2017)	`Komacek2017PhaseCurvePopulationModel`	[illegible]
Fig 2: Hotspot Peak Offset	Komacek et al. (2017)	`Komacek2017PhaseCurvePopulationModel`	[illegible]

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.